MERIDIAN CAPITAL TERMINAL

Designing *Without Thought*

11 interaction patterns inspired by Naoto Fukasawa's philosophy — where motion is purposeful but invisible, and the interface simply feels alive.

C# · WINUI 3 · XAML · UNO PLATFORM

PHILOSOPHY

The object should feel alive and aware of its own content

Design dissolving in behavior
The best interaction is one the user never notices

Without thought
Actions feel instinctive — the interface anticipates

The outline of things
Every animation must serve the data, not decorate it

GPU CONSTRAINT

Uno Platform hardware-accelerates only Opacity and RenderTransform. Every interaction in this system is designed within this boundary.

ARCHITECTURE

Six principles, eleven interactions

- The object knows what time it is — 01 11
- Confirm without congratulating — 02
- Reduce, then reduce again — 04 07
- The absence teaches you… — 05 06
- The hand knows before the eye — 03 08 09
- One material, one gesture — 10

`Storyboard` `CompositeTransform` `ShadowContainer` `SKXamlCanvas` `VisualStateManager` `HueRotationEffect`

01

Market Breathing

The object knows what time it is

Card borders pulse with a barely-visible glow during market hours, then go completely still after close.

PERFORMANCE

$142,847

5s cycle · 3% opacity · market hours only

ShadowContainer with animated BlurRadius (0–6) and Opacity (0–0.04). Gated by DispatcherTimer checking market hours.

ShadowContainer | Storyboard Forever | DispatcherTimer

02

Chart Data Settle

Confirm without congratulating

The chart line overshoots by 2–4% on Y when it finishes drawing, then springs back like an analog gauge needle.

Three-phase CompositeTransform.ScaleY keyframes: snap 1.04, ease through 0.99, rest at 1.0 with BackEase. GPU-bound.

CompositeTransform.ScaleY | KeyFrames | BackEase EaseOut

03

Ink Spread

The hand knows before the eye

Timeframe button fill radiates from the exact click point instead of appearing uniformly.

PointerPressed spawns an Ellipse at click coordinates. Scale 0–3 over 300ms, Opacity 1–0 over 400ms. GPU-bound.

PointerPressed | Canvas ClipToBounds | Scale + Opacity

04

Weight Whisper

Reduce, then reduce again

Hovering a holding subtly shifts the chart gradient density proportional to portfolio weight.

gradient α tracks portfolio weight

SpringAnimator interpolates gradient alpha inside SKXamlCanvas.OnPaintSurface.
AAPL 24% = dense. JPM 8% = faint.

SKXamlCanvas · SpringAnimator · PointerEntered

05

Silence on Leave

The absence teaches you what was there

Card content dips to 92% opacity for 600ms when the cursor exits. The chart exhales.

PointerExited fires DoubleAnimationUsingKeyFrames: 200ms delay, dip to 0.92, recover over 400ms. GPU-bound.

Opacity KeyFrames · 200ms delay · CubicEase

06

Empty State Gravity

The absence teaches you what was there

Zero search results gain an inward shadow and desaturated border.
Emptiness has physical weight.

ShadowContainer IsInner=True. VisualStateManager toggles HasResults/Empty.
VisualTransition 500ms.

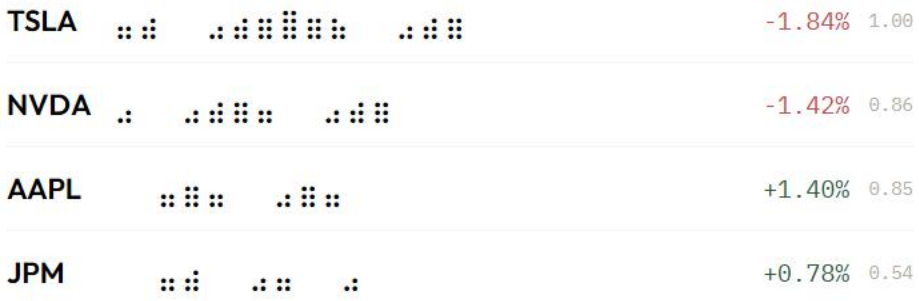
ShadowContainer Inner | VisualStateManager | VisualTransition

07

Data-Driven Braille

Reduce, then reduce again

Braille indicators tethered to actual percentage change. Volatile stocks pulse fast, calm ones are nearly still.



intensity = min(1.0, |pct|/2 + 0.15). DispatcherTimer 60fps drives SKXamlCanvas per visible row.

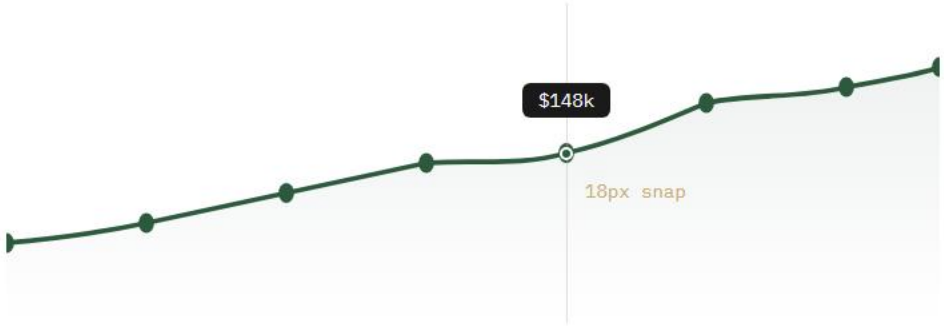
SKXamlCanvas | DispatcherTimer 16ms | f(|pctChange|)

08

Tooltip Magnetism

The hand knows before the eye

The chart crosshair snaps to data points with acceleration and eases lazily between them.



lerp factor 0.7 when snapped within 18px, 0.4 when free. All in one SKXamlCanvas draw call.

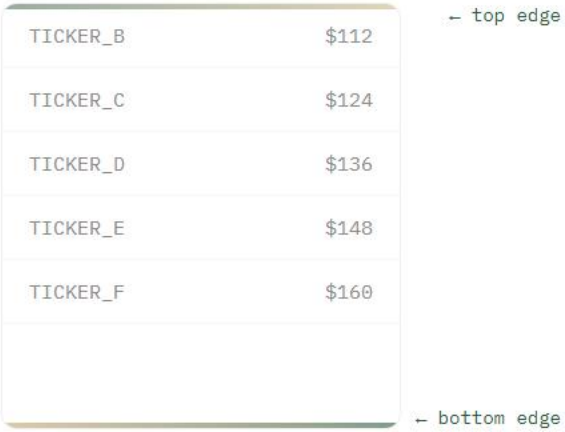
SKXamlCanvas | lerp interpolation | Snap 18px

09

Scroll Anticipation

The hand knows before the eye

Accent-colored edge bars appear at scroll boundaries. The list has edges you can feel.



ScrollViewer.ViewChanged checks VerticalOffset. Four overlay Borders via VisualStateManager. GPU-bound Opacity.

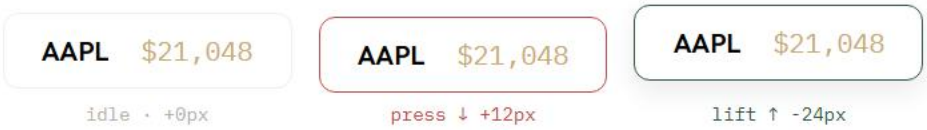
ScrollViewer | VisualStateManager | Opacity GPU

10

Weighted Paper

One material, one gesture

Hover lift uses spring physics. Cards press down before rising, giving every motion mass.



TranslateY keyframes: 0 to +3 (80ms press), +3 to -6 (500ms ElasticEase). Return 0 over 400ms.

TranslateY | ElasticEase | ShadowContainer

11

Session Warmth

The object knows what time it is

Ambient orbs drift warmer in hue the longer you sit. A patina of use, capping at 25 degrees.

minute 1 · 0°

minute 20 · 25°

Composition HueRotationEffect on orb Visual. DispatcherTimer 10s. Compositor thread, zero UI cost.

HueRotationEffect | DispatcherTimer 10s | Cap 25 deg

TAKEAWAY

Ship the invisible ones first

The interactions that add the most life are the ones users never consciously notice. Start with what makes the dashboard feel different, not look different.

P1	07	**Data-Driven Braille**	Decoration → information
P2	01	**Market Breathing**	Time-awareness, zero UI
P3	05	**Silence on Leave**	One storyboard, max subtlety
P4	10	**Weighted Paper**	Unified motion vocabulary
P5	02	**Chart Data Settle**	Analog gauge satisfaction

Design dissolves in behavior.

MERIDIAN TERMINAL V4.1 · UNO PLATFORM